

Instructions:

You will work on these exercises during the practical sessions in the fifth week of the course.

Below is an indication of the exercises that we will cover (at least) in each session. We encourage you try them before the session.

- First session (Wednesday/Thursday): exercises 1, 2.
- Second session (Friday): exercise 4.

Exercise 1: Constrained optimization

Consider the constrained optimization problem

$$\min_{\mathbf{x} \in \mathbb{R}^2} x_1 x_2 \text{ such that } x_1^2 + x_2^2 = 1.$$

Let's solve the problem via the method of Lagrange multipliers:

- (a) Write the constrained optimization problem in the form

$$\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}) \text{ subject to } g(\mathbf{x}) = c.$$

Solution:

$$f(\mathbf{x}) = x_1 x_2, \quad g(\mathbf{x}) = x_1^2 + x_2^2.$$

- (b) Write the Lagrangian $L(\mathbf{x}, \lambda)$.

Solution:

$$L(\mathbf{x}, \lambda) = f(\mathbf{x}) + \lambda(g(\mathbf{x}) - c) = x_1 x_2 + \lambda(x_1^2 + x_2^2 - 1).$$

- (c) Write down the stationary conditions for the Lagrangian.

Solution:

$$\frac{\partial L}{\partial x_1} = x_2 + 2\lambda x_1, \quad \frac{\partial L}{\partial x_2} = x_1 + 2\lambda x_2, \quad \frac{\partial L}{\partial \lambda} = x_1^2 + x_2^2 - 1.$$

We set the derivatives equal to zero to obtain the stationary conditions:

$$\frac{\partial L}{\partial x_1} = 0, \quad \frac{\partial L}{\partial x_2} = 0, \quad \frac{\partial L}{\partial \lambda} = 0,$$

obtaining:

$$x_2 + 2\lambda x_1 = 0, \tag{1}$$

$$x_1 + 2\lambda x_2 = 0, \tag{2}$$

$$x_1^2 + x_2^2 = 1. \tag{3}$$

- (d) Find all the stationary points $(\mathbf{x}^*, \lambda^*)$.

Solution:

Multiplying equation (1) by x_2 , equation (2) by $-x_1$, and adding the resulting equations gives

$$x_1^2 = x_2^2.$$

Substituting this into equation (3) yields

$$2x_1^2 = 1 \iff x_1 = \pm \frac{1}{\sqrt{2}}.$$

Therefore, there are four solutions of the Lagrange multiplier equations:

$$(x_1^*, x_2^*, \lambda^*) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, -\frac{1}{2} \right),$$

$$(x_1^*, x_2^*, \lambda^*) = \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, -\frac{1}{2} \right),$$

$$(x_1^*, x_2^*, \lambda^*) = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, \frac{1}{2} \right),$$

$$(x_1^*, x_2^*, \lambda^*) = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{2} \right).$$

(e) Which of the stationary points that you found are actually minimums?

Solution:

We evaluate the objective at the four stationary points:

$$f\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \frac{1}{2}, \quad f\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right) = \frac{1}{2},$$

$$f\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right) = -\frac{1}{2}, \quad f\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = -\frac{1}{2}.$$

Hence the global minima are

$$\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right) \quad \text{and} \quad \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right).$$

Exercise 2: Mixture models

Consider a data distribution whose underlying generating process is a mixture of Poisson distributions, but we do not know the parameters of the mixture model. In this question, you are asked to derive the updated equations for the general Poisson mixture model. The Poisson distribution is:

$$p(x|\lambda) = \frac{1}{x!} \lambda^x \exp(-\lambda)$$

where $x = 0, 1, 2, \dots$ (non-negative integers), $\lambda > 0$ is the 'rate' of the data; the expected value of x is λ . A mixture representation assumes the following:

$$p(x_n) = \sum_{k=1}^K \pi_k p(x_n|\lambda_k)$$

where $p(x_n|\lambda_k)$ is a Poisson distribution with rate λ_k and x_n is a single data observation. To answer the following questions assume we are given a dataset $\{x_1, x_2, \dots, x_N\}$. Make sure that the

constraint $\sum_k \pi_k = 1$ is satisfied (i.e. think of the log-likelihood or log-joint as f (an objective to maximize) and $\sum_k \pi_k - 1 = 0$ as $g = 0$ (a constraint that must hold)).

- (a) Write down the likelihood (as usual) for the data set in terms of $\{x_1, x_2, \dots, x_N\}$, $\{\pi_k\}$ and $\{\lambda_k\}$.

Solution:

The likelihood of using our mixture representation can be written as:

$$\mathcal{L} \stackrel{\text{i.i.d.}}{=} \prod_{n=1}^N \sum_{k=1}^K \pi_k \frac{1}{x_n!} \lambda_k^{x_n} \exp(-\lambda_k)$$

- (b) Write down the log-likelihood (as usual) for the data set in terms of $\{x_1, x_2, \dots, x_N\}$, $\{\pi_k\}$, $\{\lambda_k\}$.

Solution:

The log-likelihood:

$$\log(\mathcal{L}) = \sum_{n=1}^N \log \sum_{k=1}^K \pi_k \frac{1}{x_n!} \lambda_k^{x_n} \exp(-\lambda_k)$$

- (c) Let us consider the contribution of each of the Poisson components as their *responsibility* in the generating process, denoting them r_{nk} (i.e. the contribution of the k th- poisson distribution for the n th datapoint).

This responsibilities are actually the posterior probabilities given by the following expression $p(C_k | \mathbf{x}_n) = r_{nk}$.

Find the expression for the responsibilities r_{nk} .

Solution:

By developing the Bayes theorem for the posterior:

$$r_{nk} = \frac{\pi_k p(x_n | \lambda_k)}{\sum_l \pi_l p(x_n | \lambda_l)}$$

- (d) Find the expression for λ_k that maximizes the log-likelihood given fixed responsibilities $\{r_{nk}\}_{n=1}^N$.

Solution:

This is solved by setting $\frac{\partial \log \mathcal{L}}{\partial \lambda_k} = 0$, write it in terms of r_{nk} and then solve for λ_k . Let us first compute $\frac{\partial \log \mathcal{L}}{\partial \lambda_k}$:

$$\begin{aligned} \frac{\partial \log \mathcal{L}}{\partial \lambda_k} &= \sum_{n=1}^N \frac{1}{\sum_l \pi_l p(x_n | \lambda_l)} \left(\pi_k \frac{x_n}{\lambda_k} p(x_n | \lambda_k) - \pi_k p(x_n | \lambda_k) \right) \\ &= \sum_{n=1}^N \frac{\pi_k p(x_n | \lambda_k)}{\sum_l \pi_l p(x_n | \lambda_l)} \left(\frac{x_n}{\lambda_k} - 1 \right), \end{aligned} \quad (1)$$

where we compute $\partial p(x_n | \lambda_k) / \partial \lambda_k = p(x_n | \lambda_k) (x_n \lambda_k^{-1} - 1)$ using the fact that $p(x_n | \lambda_k)$ has a product of two terms ($\lambda_k^{x_n}$ and $\exp(-\lambda_k)$) that depend on λ so we used the product rule of differentiation to obtain (1).

Next we identify all instances of $r_{nk} = \frac{\pi_k p(x_n|\lambda_k)}{\sum_l \pi_l p(x_n|\lambda_l)}$:

$$\frac{\partial \log L}{\partial \lambda_k} = \sum_{n=1}^N r_{nk} \left(\frac{x_n}{\lambda_k} - 1 \right).$$

Now we solve $\frac{\partial \log \mathcal{L}}{\partial \lambda_k} = 0$ for λ_k

$$\begin{aligned} \sum_{n=1}^N r_{nk} \left(\frac{x_n}{\lambda_k} - 1 \right) &= 0 \\ \Leftrightarrow \sum_n r_{nk} \frac{x_n}{\lambda_k} &= \sum_n r_{nk} \\ \Leftrightarrow \lambda_k &= \frac{\sum_n r_{nk} x_n}{\sum_{n=1}^N r_{nk}}. \end{aligned}$$

So $\lambda_k = \sum_n r_{nk} x_n / N_k$ with $N_k = \sum_{n=1}^N r_{nk}$.

- (e) Find the expression for π_k that maximizes the log-likelihood given fixed responsibilities $\{r_{nk}\}_{n=1}^N$.

Solution:

Note that the solution for π_k should satisfy the constraint $\sum_k \pi_k = 1$. Hence we obtain the solution via the method of Lagrange multipliers. So first we write down the Lagrangian, then we compute the stationary points w.r.t π_k and Lagrange multiplier β . The Langrangian is given as follows:

$$L = \sum_{n=1}^N \log \sum_{k=1}^K \pi_k \frac{1}{x_n!} \lambda_k^{x_n} \exp(-\lambda_k) + \beta \left(\sum_k \pi_k - 1 \right).$$

Now we find the stationary point for π_k (i.e. solve $\frac{\partial L}{\partial \pi_k}$). The derivative of the Langrangian w.r.t π_k is given by

$$\begin{aligned} \frac{\partial L}{\partial \pi_k} &= \sum_n \frac{p(x_n|\lambda_k)}{\sum_l \pi_l p(x_n|\lambda_l)} + \beta \\ &= \sum_n \frac{r_{nk}}{\pi_k} + \beta. \end{aligned}$$

Let us set it to zero and solve for π_k :

$$\begin{aligned} \sum_n \frac{r_{nk}}{\pi_k} + \beta &= 0 \\ \Leftrightarrow \pi_k &= - \sum_n \frac{r_{nk}}{\beta}. \end{aligned} \tag{2}$$

The stationary point for β is obtained by solving

$$\begin{aligned} \frac{\partial L}{\partial \beta} &= 0 \\ \Leftrightarrow \sum_k \pi_k &= 1, \end{aligned}$$

in which we substitute the result of (2) to obtain:

$$\begin{aligned} \sum_k - \sum_n \frac{r_{nk}}{\beta} &= 1, \\ \Leftrightarrow \beta &= - \sum_n 1 = -N, \end{aligned}$$

where we note that $\sum_k r_{nk} = 1$. Now that we have an expression for β we get our final result from (2) as $\pi_k = \frac{1}{N} \sum_{n=1}^N r_{nk}$.

- (f) Now assume priors for π_k and λ_k , $p(\lambda_k|a, b) = \mathcal{G}(\lambda_k|a, b)$ (a Gamma prior) and $p(\pi_1, \dots, \pi_k) = \mathcal{D}(\pi_1, \dots, \pi_k|\alpha/K, \dots, \alpha/K)$ (a Dirichlet distribution) respectively. These distributions are defined in the appendix of Bishop. Write down the log-joint distribution:

$$\log p(x_1, \dots, x_N, \{\pi_k\}, \{\lambda_k\}|a, b, \alpha, K).$$

Solution:

$$\tilde{\mathcal{L}} = \log \mathcal{L} + \sum_k \log \mathcal{G}(\lambda_k|a, b) + \log \mathcal{D}(\{\pi_k\}|\alpha, K)$$

whatever is not depending in λ_k can be considered as constant C

$$\tilde{\mathcal{L}} = \log \mathcal{L} + \sum_k (a-1) \log \lambda_k - b \lambda_k + \sum_k (\alpha/K - 1) \log \pi_k + C$$

- (g) Find the expression for λ_k that maximizes the log-joint given fixed responsibilities $\{r_{nk}\}_{n=1}^N$.

Solution:

Same recipe as before but now applied to $\tilde{\mathcal{L}}$:

$$\begin{aligned} \frac{\partial \tilde{\mathcal{L}}}{\partial \lambda_k} &= \sum_n r_{nk} (x_n \lambda_k^{-1} - 1) + (a-1) \lambda_k^{-1} - b = 0 \\ \Leftrightarrow \lambda_k^{-1} (\sum_n r_{nk} x_n + a - 1) - (\sum_n r_{nk} + b) &= 0 \\ \Leftrightarrow \lambda_k &= \frac{\sum_n r_{nk} x_n + a - 1}{N_k + b}. \end{aligned}$$

- (h) Find the expression for π_k that maximizes the log-joint given fixed responsibilities $\{r_{nk}\}_{n=1}^N$.

Solution:

In the following $\tilde{L}$ denotes the Lagrangian (previous $\tilde{\mathcal{L}} + \beta(\sum_k \pi_k - 1)$). Find stationary point w.r.t π_k :

$$\begin{aligned} \frac{\partial \tilde{L}}{\partial \pi_k} &= \sum_n \frac{p(x_n|\lambda_k)}{\sum_l \pi_l p(x_n|\lambda_l)} + \beta + (\alpha/K - 1) \pi_k^{-1} = 0 \\ \Leftrightarrow \sum_n r_{nk} + \pi_k \beta + \alpha/K - 1 &= 0 \end{aligned}$$

$$\Leftrightarrow \pi_k = -\frac{\sum_n r_{nk} + \alpha/K - 1}{\beta}$$

Stationary point for β :

$$\sum_k \pi_k = 1$$

$$\Leftrightarrow -\frac{\sum_k \sum_n r_{nk} + \alpha/K - 1}{\beta} = 1$$

$$\Leftrightarrow \beta = -(N + \alpha - K).$$

Hence $\pi_k = \frac{\sum_n r_{nk} + \alpha/K - 1}{N + \alpha - K}.$

- (i) Write down an iterative algorithm using the above update equations (similar to the ones derived in class for the Mixture of Gaussians); include initialization and convergence check steps.

Solution:

- (a) Randomly assign data to K clusters (i.e. set hard values for r_{nk}). Compute π_k and λ_k from the initial assignments.
- (b) Update (soft) r_{nk} (E-step).
 - i. Update π_k using MLE or MAP estimates from above (M-step for π_k).
 - ii. Update λ_k using MLE or MAP estimates from above (M-step for λ_k).
 - iii. Compute $\mathcal{L}$ (or $\tilde{\mathcal{L}}$) and repeat E and M steps until $\Delta\mathcal{L} < \epsilon$ (or $\Delta\tilde{\mathcal{L}} < \epsilon$).

Exercise 3: Mixture of Experts in the Mushroom Kingdom

Exam question

Mamma mia! The evil Bowser has kidnapped Princess Peach again. Mario is about to start his journey to Bowser's castle, but, this time, he wants to apply some of the knowledge he acquired during the ML1 course to make his path a bit easier. More specifically, he wants to model the time between piranha plants coming out of their pipes to gain an advantage over them.

Mario has found some old files in Princess Peach's castle that state the following:

- There are three types of piranha plants: *regular*, *fire*, and *ice*.
- For a specific type of piranha plant, the time between pop-outs follows an exponential distribution, which has the following density:

$$p(x \mid \lambda) = \lambda e^{-\lambda x}, \quad x \geq 0, \quad \lambda > 0.$$

- He also found a data file with a list of pop-up times of piranha plants, but the plant types are not specified.

- (a) Given the information he gathered, *Mario* decided to model the time between appearances of piranha plants of any type with a mixture of exponential distributions.

Do/answer the following:

- (i) Write down the probability density function for such model.

- (ii) Clearly define each of the random variables involved.
 (iii) How many trainable parameters does the model have?

Solution:

Let x denote the time between two consecutive pop-outs and let z denote the a-priori unknown plant type, where

$$z \in \{\mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3\}.$$

Writing $\pi_l = p(z = \mathcal{C}_l)$ and using a rate λ_l for plant type $\mathcal{C}_l$, the mixture density is

$$p(x) = \sum_{l=1}^3 p(x | z = \mathcal{C}_l) p(z = \mathcal{C}_l) = \sum_{l=1}^3 \pi_l \lambda_l e^{-\lambda_l x}.$$

The trainable parameters are the three mixing probabilities π_1, π_2, π_3 and the three rates $\lambda_1, \lambda_2, \lambda_3$, giving six parameters in total.

- (b) Write the log-likelihood function for a given dataset of samples $\mathcal{D} = \{x_1, x_2, \dots, x_N\}$ for this model.

Solution:

Assuming that the observations are i.i.d.,

$$p(\mathcal{D}) = \prod_{n=1}^N p(x_n) = \prod_{n=1}^N \sum_{l=1}^3 \pi_l \lambda_l e^{-\lambda_l x_n}.$$

The log-likelihood is therefore

$$\log p(\mathcal{D}) = \sum_{n=1}^N \log \left(\sum_{l=1}^3 \pi_l \lambda_l e^{-\lambda_l x_n} \right).$$

- (c) Write down the expression for the responsibility of expert k on datapoint n , i.e., $r_{nk} = p(z = \mathcal{C}_k | x_n)$.

Solution:

By Bayes' rule,

$$r_{nk} = p(z = \mathcal{C}_k | x_n) = \frac{p(x_n | z = \mathcal{C}_k) p(z = \mathcal{C}_k)}{p(x_n)}.$$

Expanding the evidence term gives

$$r_{nk} = \frac{\pi_k \lambda_k e^{-\lambda_k x_n}}{\sum_{l=1}^3 \pi_l \lambda_l e^{-\lambda_l x_n}}.$$

- (d) Mario now wants to apply the Expectation-Maximization algorithm to learn optimal parameters for the model. Formulate the optimization problems that need to be solved to update the parameters in the Maximization step.

Note: You do not need to compute the analytical expressions for the parameter updates, but only specify the optimization problem that needs to be solved to obtain them.

Solution:

The mixing probabilities are updated by solving

$$\pi_k \leftarrow \arg \max_{\pi_k} \log p(\mathbf{x} \mid \boldsymbol{\lambda}, \boldsymbol{\pi}) \text{ subject to } \pi_1 + \pi_2 + \pi_3 = 1.$$

For each expert k , the rate is updated by solving

$$\lambda_k \leftarrow \arg \max_{\lambda_k} \log p(\mathbf{x} \mid \boldsymbol{\lambda}, \boldsymbol{\pi}).$$

(When solving both optimization problems the responsibilities would be fixed).

- (e) With his model trained, Mario enters a completely dark cave where he can't see anything. Suddenly, he hears a piranha plant nearby. By listening closely, he notices it pops out from its pipe every 5 seconds. How can he use his trained model to predict which type of piranha plant it is?

Note: you can assume that, after training the model, Mario has observed the resulting clusters and determined which one corresponds to each piranha plant type.

Solution:

Mario computes the responsibility of every plant type for the observation $x = 5$ and predicts the class with the highest responsibility:

$$\hat{k} = \arg \max_{k \in \{1,2,3\}} p(z = \mathcal{C}_k \mid x = 5).$$

Exercise 4: Principal component analysis

Suppose we have a data set $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$ of D -dimensional vectors, which have a zero mean for each dimension. Assume we perform a complete eigenvalue decomposition of the empirical covariance matrix $\mathbf{S} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$. You are interested in only a single projection of your data such that the variance of this projection is maximized. Let $\mathbf{u}_i$ be the direction vector of a particular projection. Assume that $\mathbf{u}_i^\top \mathbf{u}_i = 1$.

- (a) What is the projection z_{ni} of a given point $\mathbf{x}_n$ under the particular vector $\mathbf{u}_i$?

Solution:

The projection of the vector $\mathbf{x}_n$ over the vector $\mathbf{u}_i$ is given by:

$$z_{ni} = \mathbf{u}_i^\top \mathbf{x}_n$$

- (b) What is the empirical mean of the projection z_i across all points $\mathbf{x}_n$?

Solution:

The empirical mean:

$$E[z_i] = \frac{1}{N} \sum_{n=1}^N \mathbf{u}_i^\top \mathbf{x}_n = \mathbf{u}_i^\top \left(\frac{1}{N} \sum_{n=1}^N \mathbf{x}_n \right) = 0$$

- (c) What is the empirical variance of the projection z_i ? Provide your answer in terms of the empirical covariance matrix $\mathbf{S}$

Solution:

The variance of a variable z is defined as:

$$V[z_i] = E[(z_i - \bar{z}_i)^2]$$

Hence, in our case:

$$\begin{aligned} V[z_i] &= \frac{1}{N} \sum_{n=1}^N (\mathbf{u}_i^T \mathbf{x}_n)(\mathbf{u}_i^T \mathbf{x}_n)^T \\ &= \frac{1}{N} \sum_{n=1}^N \mathbf{u}_i^T \mathbf{x}_n \mathbf{x}_n^T \mathbf{u}_i \\ &= \mathbf{u}_i^T \mathbf{S} \mathbf{u}_i \end{aligned}$$

- (d) Replace $\mathbf{S}$ with its eigenvalue decomposition and simplify the aforementioned expression. What is the variance now?

Solution:

$$V[z_i] = \mathbf{u}_i^T \mathbf{S} \mathbf{u}_i = \mathbf{u}_i^T \mathbf{U} \mathbf{\Lambda} \mathbf{U}^T \mathbf{u}_i$$

Since $\mathbf{u}_i^T \mathbf{u}_i = 1$ and $\mathbf{u}_i^T \mathbf{u}_j = 0$ then we can re-write:

$$= \mathbf{e}_i^T \mathbf{\Lambda} \mathbf{e}_i = \lambda_i$$

with $\mathbf{e}_i$ to be a vector with zeros except the position with index i .

- (e) Suppose that you are interested in reducing the dimensionality from D to K , such that 99% of the variance is maintained. How can you select an appropriate K ?

Solution:

We need to sort eigenvalues in descending order. Then by picking K largest eigenvalues using the following formula:

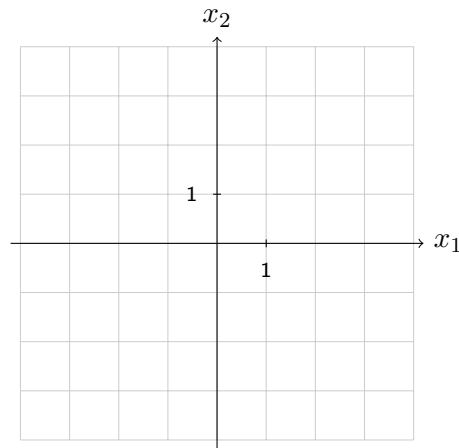
$$\frac{\sum_{i=1}^{K-1} \lambda_i}{\sum_{i=1}^D \lambda_i} < 0.99 \leq \frac{\sum_{i=1}^K \lambda_i}{\sum_{i=1}^D \lambda_i}$$

Exercise 5: Dimensionality reduction

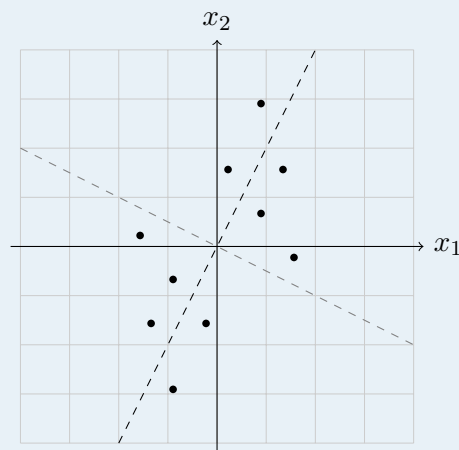
Exam question

We apply Principal Component Analysis (PCA) to a 2-dimensional dataset $\mathbf{x}$, obtaining that its first principal component is $\mathbf{u}_1 = \left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)^T$ and its second principal component is $\mathbf{u}_2 = \left(-\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)^T$.

- (a) Draw an example of a dataset $\mathbf{x}$ that satisfies this on the plot below. Include at least 10 samples.



Solution:



For questions b and c we will assume that $\mathbf{x}$ is a dataset with zero mean.

- (b) Compute the projection of $\mathbf{x}_0 = (2, 2)^\top$ onto the subspace defined by the first principal component.

Solution:

The coordinate of $\mathbf{x}_0$ along the first principal component is

$$z_0 = \mathbf{u}_1^\top \mathbf{x}_0 = \frac{1}{\sqrt{5}} 2 + \frac{2}{\sqrt{5}} 2 = \frac{6}{\sqrt{5}}.$$

- (c) If we represent the projection of $\mathbf{x}_0 = (2, 2)^\top$ on the first component in the original 2D space, what will be the reconstruction error with respect to the original datapoint?

Solution:

The coordinates of the projection in the original 2D space are:

$$\hat{\mathbf{x}}_0 = z_0 \mathbf{u}_1 = \frac{6}{\sqrt{5}} \left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right)^\top = \left(\frac{6}{5}, \frac{12}{5} \right)^\top.$$

The reconstruction residual is

$$\mathbf{x}_0 - \hat{\mathbf{x}}_0 = \begin{pmatrix} 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 6/5 \\ 12/5 \end{pmatrix} = \begin{pmatrix} 4/5 \\ -2/5 \end{pmatrix}.$$

Hence the squared reconstruction error is

$$\|\mathbf{x}_0 - \hat{\mathbf{x}}_0\|^2 = \left(\frac{4}{5}\right)^2 + \left(-\frac{2}{5}\right)^2 = \frac{4}{5}.$$

Equivalently, the Euclidean reconstruction error is $2/\sqrt{5}$.